Sprint 1 Plan – Course Registration System (VCU Summer 2026)

1. Product Vision

The vision of this project is to develop a simple, user-friendly course registration system for the 2026 summer session at VCU that allows students to easily search for courses, view available sessions, and enroll efficiently. The system will provide accurate, real-time information for students, instructors, and administrators, improving accessibility, reducing manual errors, and enhancing the overall registration experience.

2. Product Backlog

User Story	Description	Acceptance Criteria	Priority	Estimate
US1	Search courses by department	Displays correct courses	High	Medium
US2	View course details	Shows course info	High	Medium
US3	Enroll in course	Updates count, no duplicates	High	High
US4	View enrolled classes	Displays schedule	Medium	Low
US5	Add courses/sessions	Data saved correctly	High	High
US6	View instructors	Displays instructor info	Medium	Low
US7	Prevent over-enrollment	Stops at max capacity	High	Medium
US8	Prevent duplicate enrollment	Blocks duplicates	High	Medium

3. Product Roadmap

The team will manage and address the product backlog using Scrum principles:

- **Prioritization by Value and Dependency**
  High-value and foundational tasks (database design, enrollment logic) are completed first to enable future features.
- **Incremental Delivery**
  Features are built in small, functional increments across sprints rather than all at once.
- **Capacity-Based Planning**
  The team selects backlog items based on available time (6–8 hours per member per week) to ensure realistic commitments.
- **Continuous Refinement**
  The backlog will be reviewed and adjusted at the end of each sprint based on progress, challenges, and feedback.
- **Risk Reduction**
  Critical system constraints (preventing over-enrollment) are implemented early to avoid major issues later.

Sprint 1: Core database, search, enrollment –

Key Focus Areas:

- Design and implement the database schema (courses, students, sessions, instructors, departments)
- Establish relationships and constraints (e.g., one course → multiple sessions, enrollment limits)
- Develop basic backend functionality (search, course retrieval, enrollment logic)
- Build simple frontend components for searching and viewing courses
- Implement critical validation rules (no duplicate enrollment, max capacity enforcement)

Outcome:

By the end of Sprint 1, the system will support a complete basic workflow:
Admin creates data → Student searches → Student views → Student enrolls

This sprint prioritizes functionality over polish, ensuring the system works end-to-end before adding complexity.

Sprint 2: Advanced features, dashboards, AWS deployment –

Key Focus Areas:

- Enhance search capabilities (filter by instructor, course number, modality)
- Develop a student dashboard to view enrolled courses
- Develop instructor view to see assigned sessions
- Improve data validation and error handling
- Refine frontend design for better usability and clarity
- Begin preparation for AWS deployment

Outcome:

The system becomes more user-friendly and feature-complete, allowing different types of users (students**,** instructors) to interact with the system more effectively.

Sprint 3: Enhancements and testing –

Key Focus Areas:

- Perform comprehensive testing (unit testing, integration testing, edge cases)
- Fix bugs and improve system performance
- Strengthen data integrity and security controls
- Finalize AWS deployment and ensure system accessibility
- Improve UI/UX based on feedback
- Add any remaining enhancements or stretch features

Outcome:

A fully functional, stable, and deployed course registration system that meets project requirements and is ready for demonstration.

4. Sprint Goal

Build the core functionality including database, course lookup, and enrollment.

The goal of Sprint 1 is to deliver a working foundational version of the course registration system that allows:

- Admins to create and manage course/session data
- Students to search for courses by department
- Students to view course details
- Students to enroll in courses with basic validation rules

5. Sprint Backlog

US1, US2, US3, US5, US7 selected to support core functionality.

US5 (Add courses and sessions) is foundational because the system cannot function without data. Before students can search or enroll, courses, sessions, and instructors must exist in the database.

US1 (Search courses) and **US2** (View course details) provide core user experience, allowing students to explore available options and understand course requirements before making decisions.

US3 (Enroll in a course) is the primary business function of the system. Without enrollment capability, the system does not fulfill its purpose.

US7 (Prevent over-enrollment) ensures data integrity and system reliability by enforcing constraints such as maximum class capacity.

6. Sprint Tasks

US1 – Search Courses by Department

- Design and implement database schema for courses and departments
- Create SQL queries to filter courses by department ID
- Develop backend API endpoint for course search functionality
- Build a frontend component to display filtered course results
- Test search functionality with multiple departments and edge cases

US2 – View Course Details

- Create detailed course view page or component

- Retrieve course attributes (name, credits, prerequisites, REAL designation)
- Fetch and display associated sessions and instructor information
- Ensure correct formatting and readability of data
- Validate that all course information is displayed accurately

US3 – Enroll in Course

- Design and implement enrollment table in database
- Create backend logic for inserting enrollment records
- Validate that a student cannot enroll in duplicate sessions of the same course
- Update session enrollment count upon successful registration
- Display confirmation or error messages to the user
- Test enrollment flow with valid and invalid scenarios

US5 – Add Courses and Sections

- Design database schema for sessions, instructors, and course relationships
- Create backend functions to insert new courses, sessions, and instructor assignments
- Implement validation rules (required fields, correct data types, valid relationships)
- Create a simple interface or input method for adding course data (can be basic for Sprint 1)
- Ensure foreign key relationships are correctly enforced (course → session → instructor)
- Test data creation and verify that records are stored correctly

US7 – Prevent Over-Enrollment

- Implement logic to check current enrollment against maximum capacity before allowing registration
- Add conditional validation in the backend enrollment process
- Return an error message if the session is full
- Prevent database insert if max capacity is reached
- Test edge cases (e.g., last available seat, simultaneous enrollments)
- Ensure the system maintains accurate enrollment counts under all conditions

Capacity Planning

Estimated total work: ~32 hours, within team capacity of 48–64 hours.